

# GNSS Error Analysis Summary – ADCN and WAGN Stations

## 1. Introduction

Analyse GNSS errors (dN, dE, dUp) at ADCN and WAGN (2022–2024) and assess seasonal and meteorological influences, including temperature, humidity, and pressure.

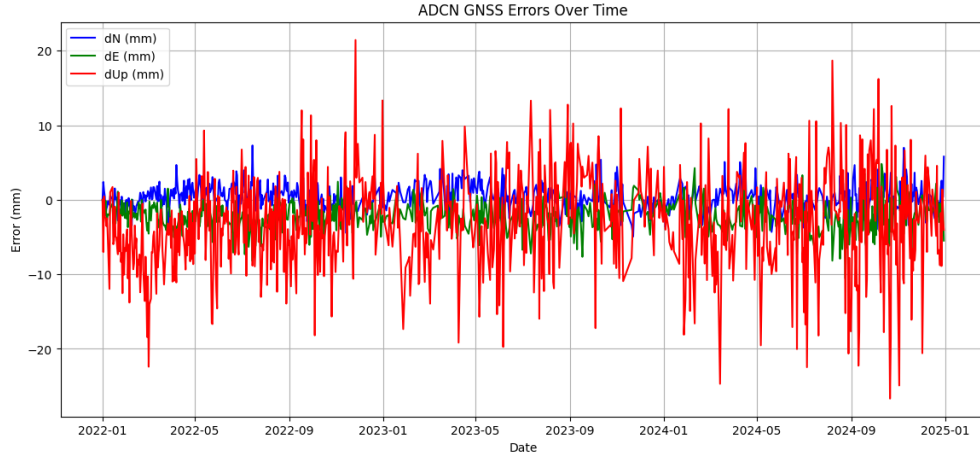
## 2. Data Cleaning and Preparation

The ADCN and WAGN datasets contained no missing values, so no imputation was needed. The DOY column was converted to a datetime format, and additional Month and Season columns were added to enable daily, monthly, and seasonal analyses. This setup allows structured temporal analysis of GNSS errors.

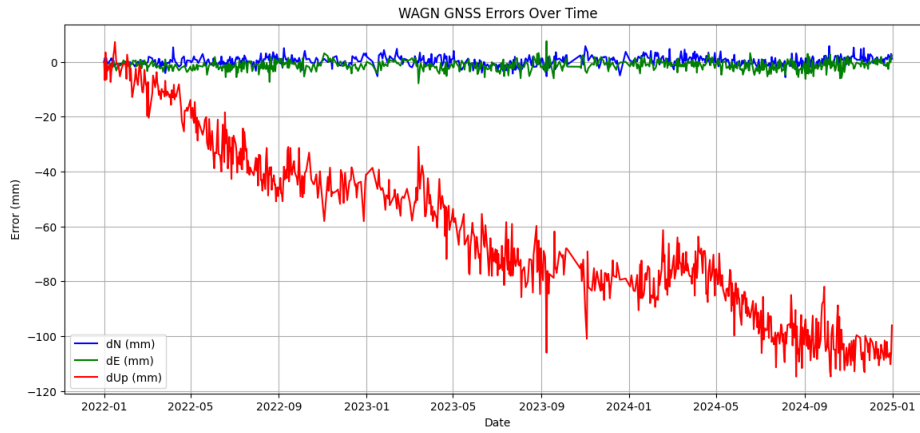
| ADCN Head with Month & Season: |                 |            |       |        |               |                   |
|--------------------------------|-----------------|------------|-------|--------|---------------|-------------------|
|                                | DOY             | dn         | de    | du     | Temp.Dry [°C] | RelHumidity [%] \ |
| 0                              | 20220101        | 0.00       | 0.00  | 0.00   | 19.69         | 79.09             |
| 1                              | 20220102        | 2.40       | -2.21 | -7.01  | 21.89         | 69.44             |
| 2                              | 20220103        | 1.24       | -1.81 | 0.46   | 22.61         | 70.73             |
| 3                              | 20220104        | 0.44       | -1.30 | -2.23  | 18.82         | 68.92             |
| 4                              | 20220105        | -2.07      | -0.14 | -2.43  | 17.12         | 72.15             |
|                                |                 |            |       |        |               |                   |
|                                | Press.QFF [hPa] | Date       | Month | Season |               |                   |
| 0                              | 1017.84         | 2022-01-01 | 1     | Winter |               |                   |
| 1                              | 1011.92         | 2022-01-02 | 1     | Winter |               |                   |
| 2                              | 1011.85         | 2022-01-03 | 1     | Winter |               |                   |
| 3                              | 1016.35         | 2022-01-04 | 1     | Winter |               |                   |
| 4                              | 1017.36         | 2022-01-05 | 1     | Winter |               |                   |
|                                |                 |            |       |        |               |                   |
| WAGN Head with Month & Season: |                 |            |       |        |               |                   |
|                                | DOY             | dn         | de    | du     | Temp.Dry [°C] | RelHumidity [%] \ |
| 0                              | 20220101        | 0.00       | 0.00  | 0.00   | 21.31         | 80.97             |
| 1                              | 20220102        | 1.45       | -1.15 | -6.70  | 22.89         | 71.60             |
| 2                              | 20220103        | 1.14       | -1.51 | 3.44   | 24.23         | 71.35             |
| 3                              | 20220104        | 1.46       | -0.36 | -6.44  | 18.34         | 84.51             |
| 4                              | 20220105        | -0.39      | -0.86 | -1.47  | 16.15         | 81.52             |
|                                |                 |            |       |        |               |                   |
|                                | Press.QFF [hPa] | Date       | Month | Season |               |                   |
| 0                              | 1018.99         | 2022-01-01 | 1     | Winter |               |                   |
| 1                              | 1014.57         | 2022-01-02 | 1     | Winter |               |                   |
| 2                              | 1012.83         | 2022-01-03 | 1     | Winter |               |                   |
| 3                              | 1017.24         | 2022-01-04 | 1     | Winter |               |                   |
| 4                              | 1018.78         | 2022-01-05 | 1     | Winter |               |                   |

Figure 1: Data with Month and Season columns

## 3. GNSS Errors over Time



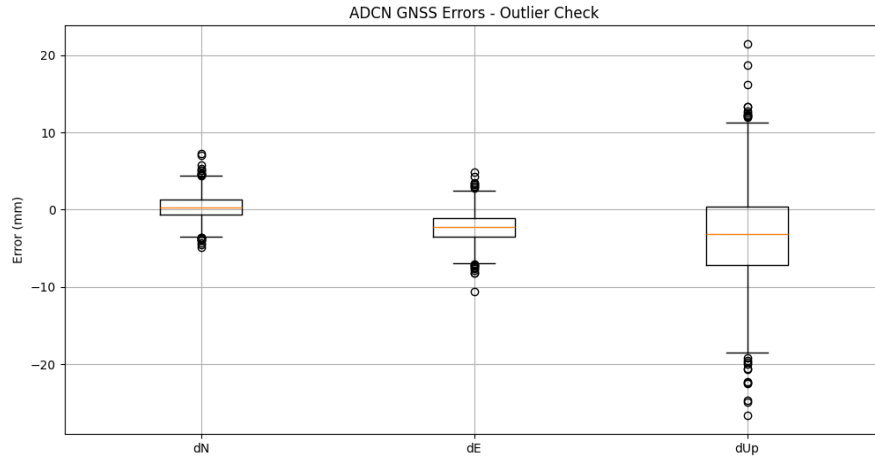
*Figure 2: ADCN GNSS Errors Over Time*



*Figure 3: WAGN GNSS Errors Over Time*

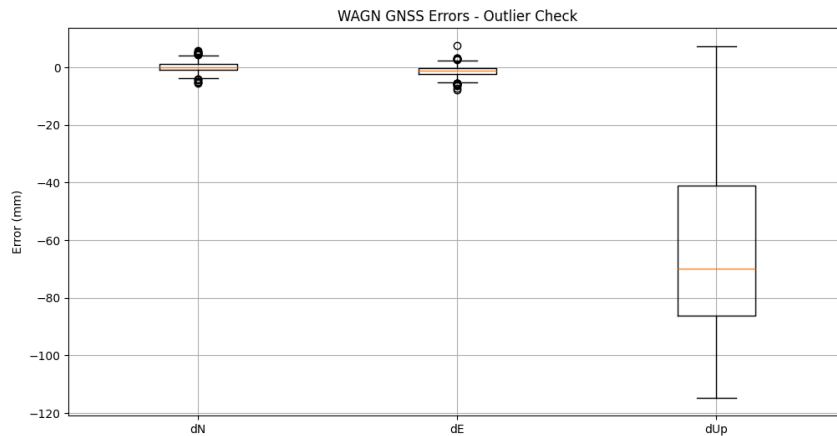
#### 4. Outliers

For ADCN, dN errors show a few points around -5 mm and 5–8 mm, while dE errors range between -10 mm and 3–5 mm. dUp exhibits the largest spread, with outliers from -27 mm to 22 mm.



*Figure 4: ADCN GNSS Outliers*

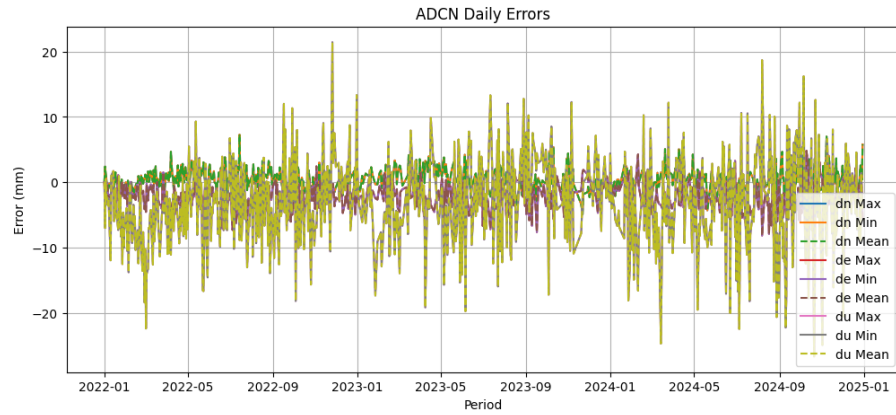
WAGN displays similar trends, but vertical errors are more pronounced; dUp centres around -70 mm with whiskers extending below -110 mm and above +10 mm. Both datasets reveal outliers beyond whiskers, indicating occasional extreme GNSS deviations, especially in the vertical component.



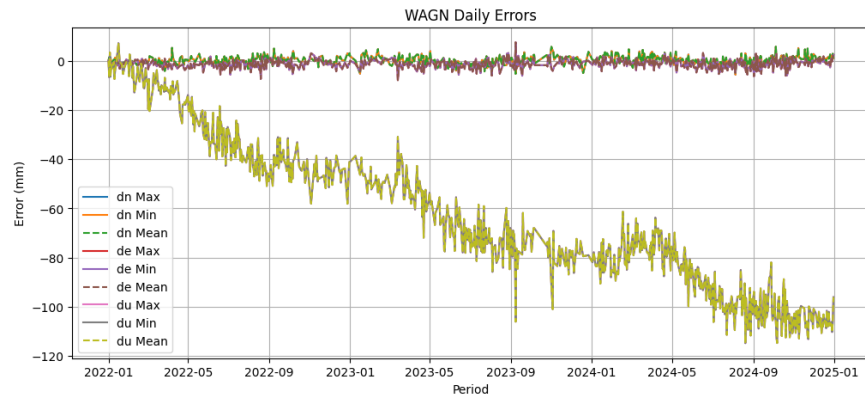
*Figure 5: WAGN GNSS Outliers*

The analysis computed daily, weekly, and monthly maximum, minimum, and mean errors for both stations, providing insight into temporal variations. ADCN shows a maximum absolute error of 26.68 mm in the vertical component (dUp), whereas WAGN experiences significantly larger vertical deviations, with a maximum of 114.81 mm. Outliers were identified using boxplots, highlighting occasional extreme errors in all three components (dN, dE, dUp), particularly in dUp. Weekly and monthly aggregation revealed patterns of recurring higher errors, suggesting seasonal or meteorological

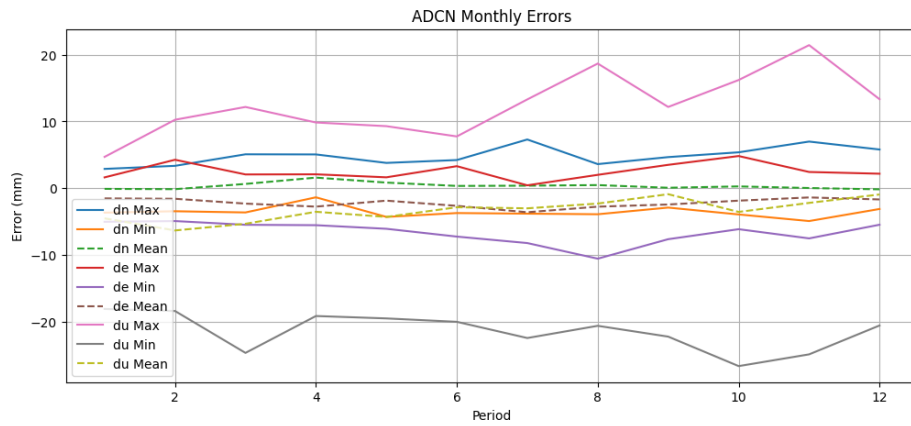
influences. These trends provide a basis for understanding GNSS error behaviour and planning corrective measures.



*Figure 6: ADCN Daily Errors*



*Figure 7: WAGN Daily Errors*



*Figure 8: ADCN Monthly errors*



Figure 9: WAGN Monthly Errors

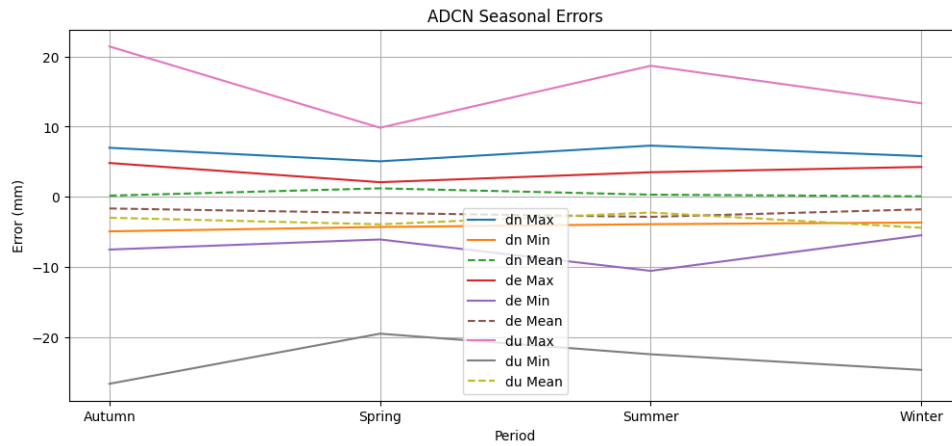


Figure 10: ADCN Seasonal Errors

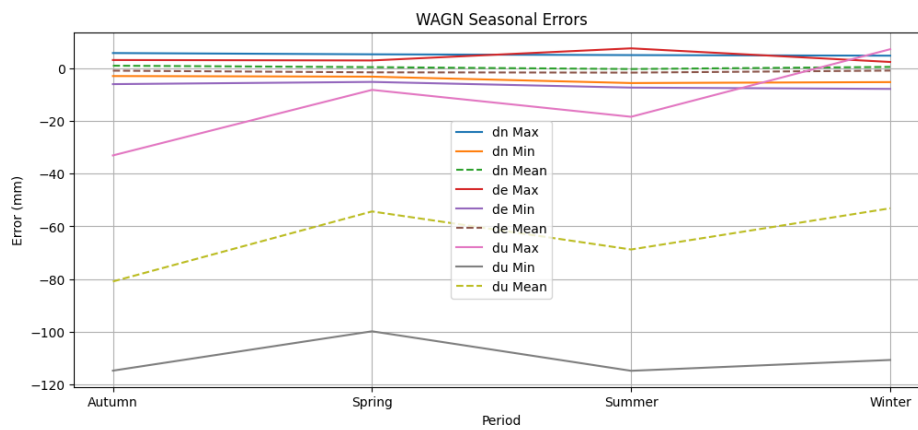


Figure 11: WAGN Seasonal Errors

## 5. Seasonal Analysis

Seasonal statistics highlight variations in GNSS positioning errors at ADCN and WAGN. For ADCN, mean errors remain relatively low across seasons, with vertical errors (dUp) showing the largest spread, particularly in Autumn (−2.99 mm, max 21.45 mm, min −26.68 mm). WAGN exhibits much larger vertical deviations, with dUp reaching extremes up to −114.81 mm and high variability in Winter (SD 34.43 mm). Representative high-error days were identified for detailed hourly analysis: ADCN shows Winter 2024-03-14 (29.70 mm total error) and Summer 2024-06-22 (30.65 mm); WAGN shows Winter 2024-12-26 (112.90 mm) and Summer 2024-08-20 (120.68 mm). These insights confirm the seasonal and station-dependent GNSS error behaviour, which is essential for modelling and correction strategies.

*Table 1: Seasonal Analysis and Representative Days*

| Station | Season | dN_mean | dE_mean | dUp_mean | dUp_max | dUp_min | Total Error (Representative Day) |
|---------|--------|---------|---------|----------|---------|---------|----------------------------------|
| ADCN    | Winter | 0.066   | -1.790  | -4.418   | 13.34   | -24.70  | 29.70 mm (2024-03-14)            |
| ADCN    | Summer | 0.308   | -2.866  | -2.243   | 18.69   | -22.47  | 30.65 mm (2024-06-22)            |
| WAGN    | Winter | 0.505   | -0.889  | -53.144  | 7.22    | -110.70 | 112.90 mm (2024-12-26)           |
| WAGN    | Summer | -0.303  | -1.663  | -68.779  | -18.42  | -114.81 | 120.68 mm (2024-08-20)           |

## 6. Statistical Metrics

The GNSS error statistics reveal the magnitude and variability of positioning deviations at ADCN and WAGN. For ADCN, horizontal errors (dN and dE) are relatively small, with mean values of 0.36 mm and −2.24 mm, standard deviations of 1.70 mm and 1.94 mm, and RMS values of 1.74 mm and 2.96 mm, respectively. Vertical errors (dUp) are larger, with a mean of −3.32 mm, SD 6.55 mm, and RMS 7.34 mm, reflecting higher

variability. At WAGN, horizontal errors remain moderate, while vertical errors (dUp) are extreme, showing a mean of  $-63.16$  mm, SD  $30.62$  mm, and RMS  $70.18$  mm. MAE and RMSE values confirm these trends, highlighting that WAGN’s vertical component dominates total error, which is crucial for error modeling and GNSS correction strategies.

ADCN Statistical Metrics:

|    | Mean      | SD       | RMS      | MAE      | RMSE     |
|----|-----------|----------|----------|----------|----------|
| dn | 0.360758  | 1.703914 | 1.740596 | 1.335373 | 1.740596 |
| de | -2.235373 | 1.939373 | 2.958571 | 2.481072 | 2.958571 |
| du | -3.322549 | 6.554791 | 7.344961 | 5.742418 | 7.344961 |

WAGN Statistical Metrics:

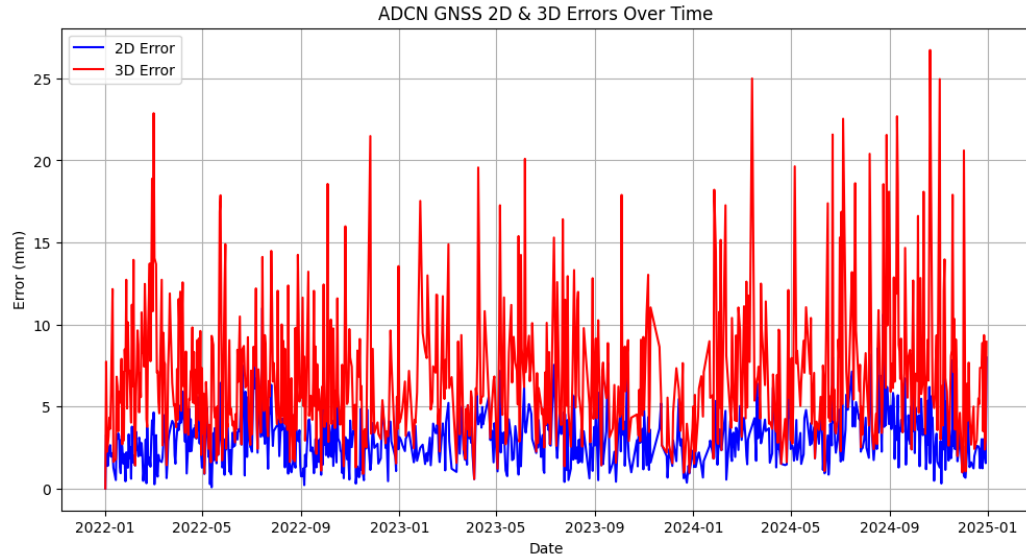
|    | Mean       | SD        | RMS       | MAE       | RMSE      |
|----|------------|-----------|-----------|-----------|-----------|
| dn | 0.240000   | 1.642898  | 1.659246  | 1.272198  | 1.659246  |
| de | -1.302802  | 1.705609  | 2.145343  | 1.698592  | 2.145343  |
| du | -63.157373 | 30.619782 | 70.179541 | 63.210697 | 70.179541 |

Figure 12: Statistical Analysis of GNSS Errors

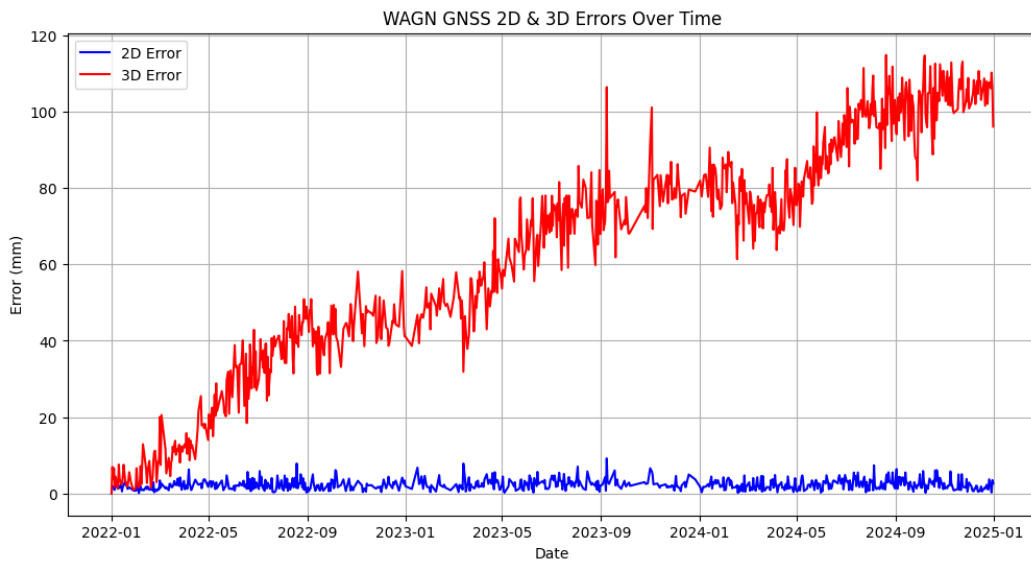


Figure 13: Statistical Comparison

## 7. 2D and 3D Errors



*Figure 14: ADCN 2D and 3D Errors Over Time*



*Figure 15: WAGN 2D and 3D Errors*

The seasonal analysis of 2D (horizontal) and 3D (total) errors highlights the variations in GNSS positioning performance at ADCN and WAGN. For ADCN, 2D errors remain moderate across seasons, with mean values ranging from 2.54 mm in winter to 3.47 mm in summer. Maximum 2D errors reach 11.0 mm in summer, while standard deviations indicate relatively low variability (1.23–1.83 mm). The 3D errors show a larger spread, with mean values between 6.60 mm (winter) and 7.23 mm (summer), and maximum



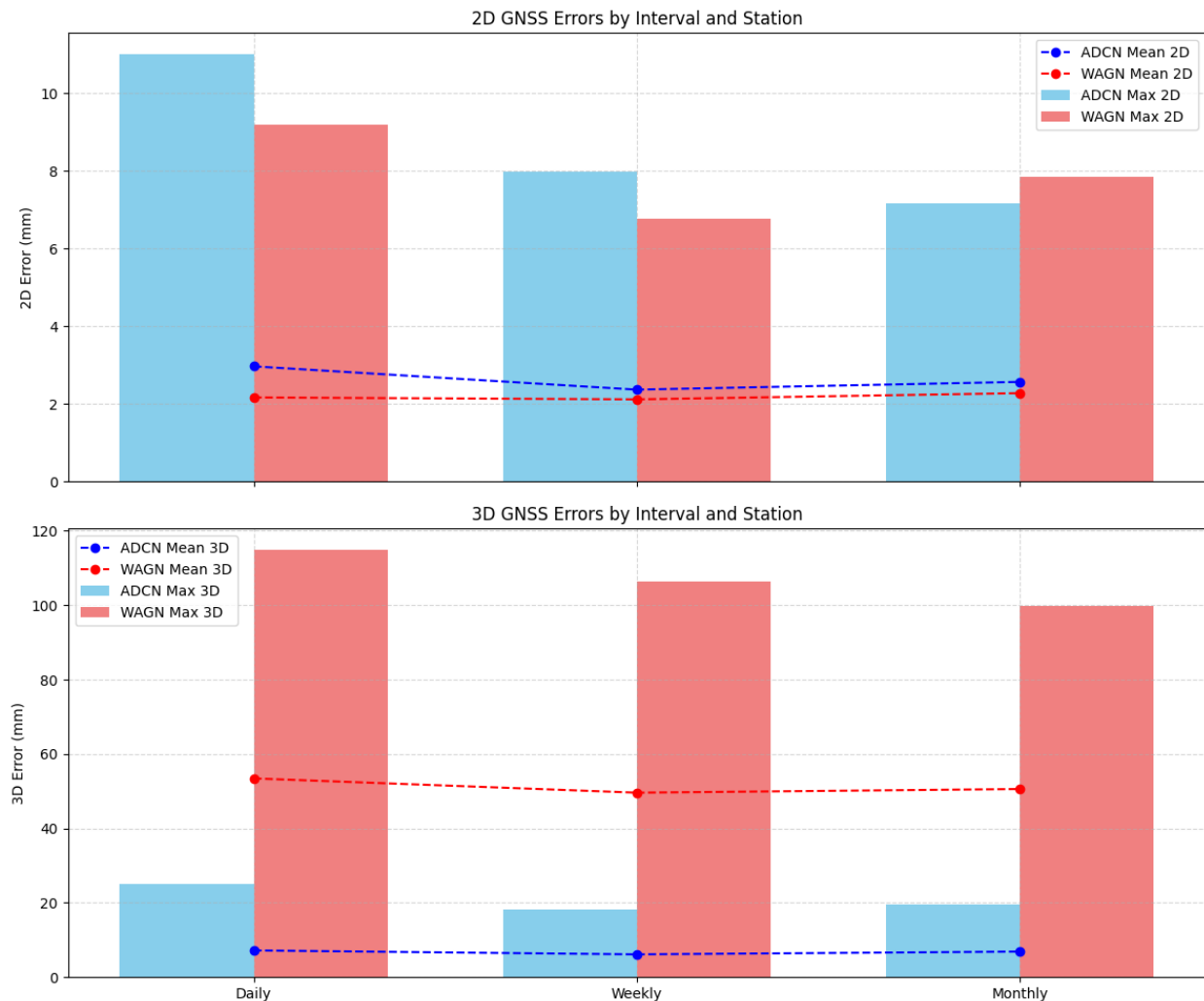
errors peaking at 26.72 mm in autumn. These results suggest seasonal variations, with summer showing slightly higher horizontal and vertical deviations.

For WAGN, 2D errors are comparable to ADCN but 3D errors are significantly larger due to extreme vertical deviations (dUp). The mean 3D error ranges from 53.45 mm in winter to 80.96 mm in autumn, with maxima exceeding 114 mm. Standard deviations are also higher (25–34 mm), reflecting pronounced variability. These findings confirm that WAGN experiences more severe vertical error excursions, especially in extreme seasons. The seasonal 2D/3D analysis emphasizes the importance of considering both horizontal and vertical components in GNSS error modeling.

*Table 2: Seasonal 2D/3D Error Stats*

| <b>Station</b> | <b>Season</b> | <b>2D Mean (mm)</b> | <b>2D Max (mm)</b> | <b>2D Min (mm)</b> | <b>2D Std (mm)</b> | <b>2D Median (mm)</b> | <b>3D Mean (mm)</b> | <b>3D Max (mm)</b> | <b>3D Min (mm)</b> | <b>3D Std (mm)</b> | <b>3D Median (mm)</b> |
|----------------|---------------|---------------------|--------------------|--------------------|--------------------|-----------------------|---------------------|--------------------|--------------------|--------------------|-----------------------|
| ADCN           | Autumn        | 2.96                | 8.31               | 0.29               | 1.61               | 2.80                  | 7.22                | 26.72              | 0.89               | 4.65               | 6.08                  |
| ADCN           | Spring        | 3.09                | 7.17               | 0.07               | 1.43               | 3.05                  | 6.64                | 19.63              | 0.57               | 3.88               | 5.71                  |
| ADCN           | Summer        | 3.47                | 11.00              | 0.21               | 1.83               | 3.43                  | 7.23                | 22.69              | 1.06               | 4.23               | 6.21                  |
| ADCN           | Winter        | 2.54                | 7.98               | 0.00               | 1.23               | 2.48                  | 6.61                | 24.99              | 0.00               | 4.05               | 5.71                  |
| WAGN           | Autumn        | 2.45                | 6.60               | 0.18               | 1.48               | 2.05                  | 80.96               | 114.77             | 33.16              | 25.92              | 83.50                 |
| WAGN           | Spring        | 2.32                | 6.27               | 0.20               | 1.25               | 2.26                  | 54.44               | 99.86              | 8.68               | 26.05              | 60.58                 |
| WAGN           | Summer        | 2.59                | 9.20               | 0.23               | 1.47               | 2.49                  | 68.85               | 114.88             | 18.46              | 26.28              | 72.83                 |
| WAGN           | Winter        | 1.99                | 7.84               | 0.00               | 1.21               | 1.79                  | 53.45               | 110.70             | 0.00               | 34.03              | 52.49                 |

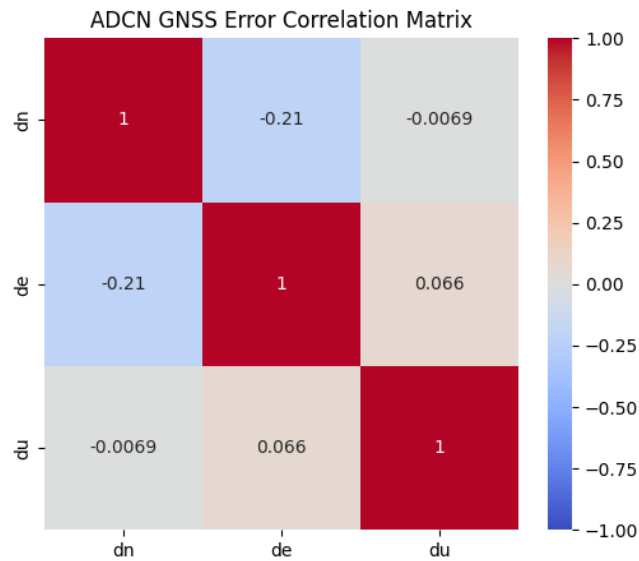
The 2D (horizontal) and 3D (spatial) errors were computed for ADCN and WAGN stations over daily, weekly, and monthly intervals. ADCN generally shows lower error magnitudes, with 2D errors ranging from 0–11 mm and 3D errors up to ~25 mm. WAGN exhibits larger vertical deviations, reflected in 3D errors exceeding 100 mm in certain weeks and months. Weekly analysis highlights periods of extreme errors, with ADCN showing weekly maxima around 8–18 mm (3D), whereas WAGN peaks above 100 mm. Monthly aggregation captures overall trends and seasonal variations. Daily statistics illustrate short-term fluctuations and occasional spikes. These comprehensive tables enable identification of high-error periods, support correlation with meteorological conditions, and facilitate further analysis for modeling GNSS performance under varying environmental conditions.



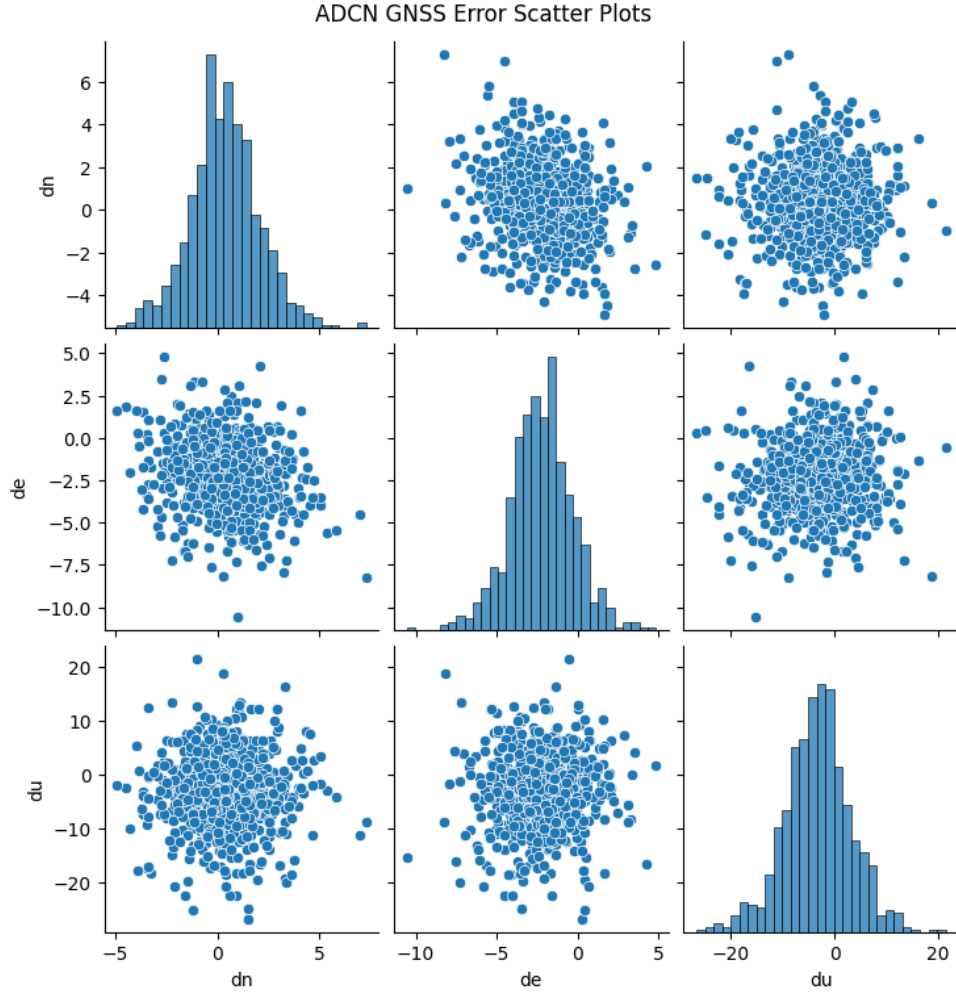
*Figure 16: ADCN and WAGN 2D and 3D Errors*

## 8. Correlation Analysis

Pearson correlation coefficients were computed to evaluate the linear relationship between the three GNSS error components (dN, dE, dUp) for both ADCN and WAGN stations. For ADCN, the correlation between dN and dE is -0.205, indicating a weak negative relationship; as northward errors increase, eastward errors tend to slightly decrease. The correlations between dN and dUp (-0.007) and between dE and dUp (0.066) are near zero, showing negligible linear association between horizontal and vertical components. This suggests that vertical errors at ADCN are largely independent of horizontal displacements.

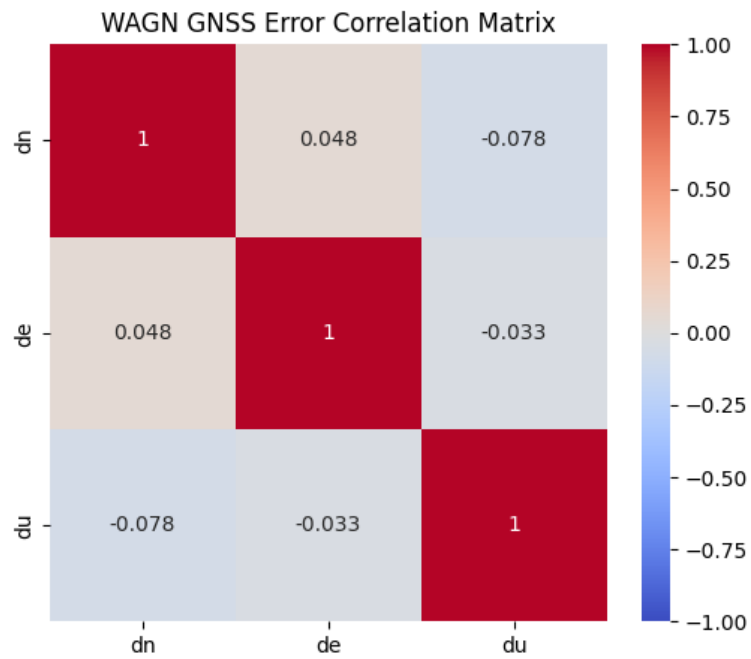


*Figure 17: ADCN Correlation Matrix*



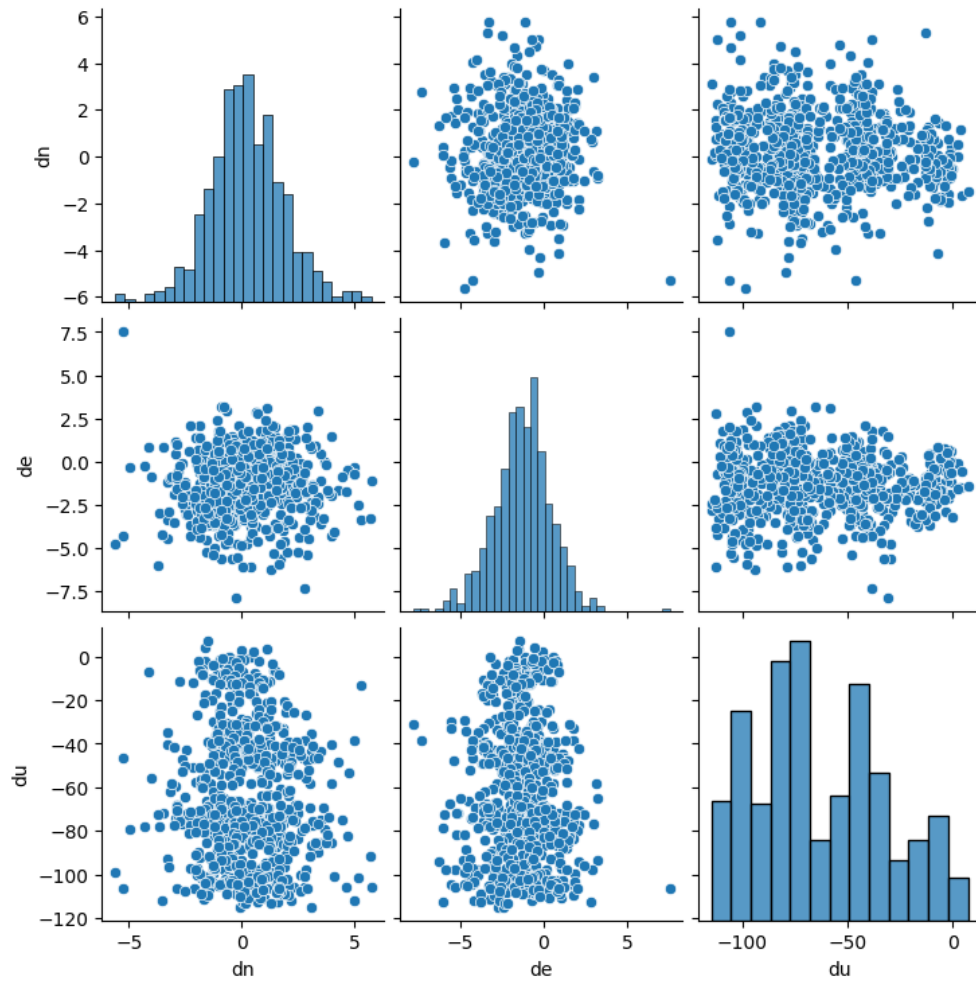
*Figure 18: ADCN Scatter Plots*

For WAGN,  $dN$  and  $dE$  exhibit a weak positive correlation of 0.048, suggesting almost no linear relationship. Similarly,  $dN$  vs  $dU$  (-0.078) and  $dE$  vs  $dU$  (-0.033) indicate very weak negative correlations, again pointing to the independence of vertical errors from horizontal errors. Compared to ADCN, WAGN shows slightly lower inter-component dependencies, reinforcing that each GNSS error component behaves largely independently.



*Figure 19: WAGN Correlation Matrix*

WAGN GNSS Error Scatter Plots



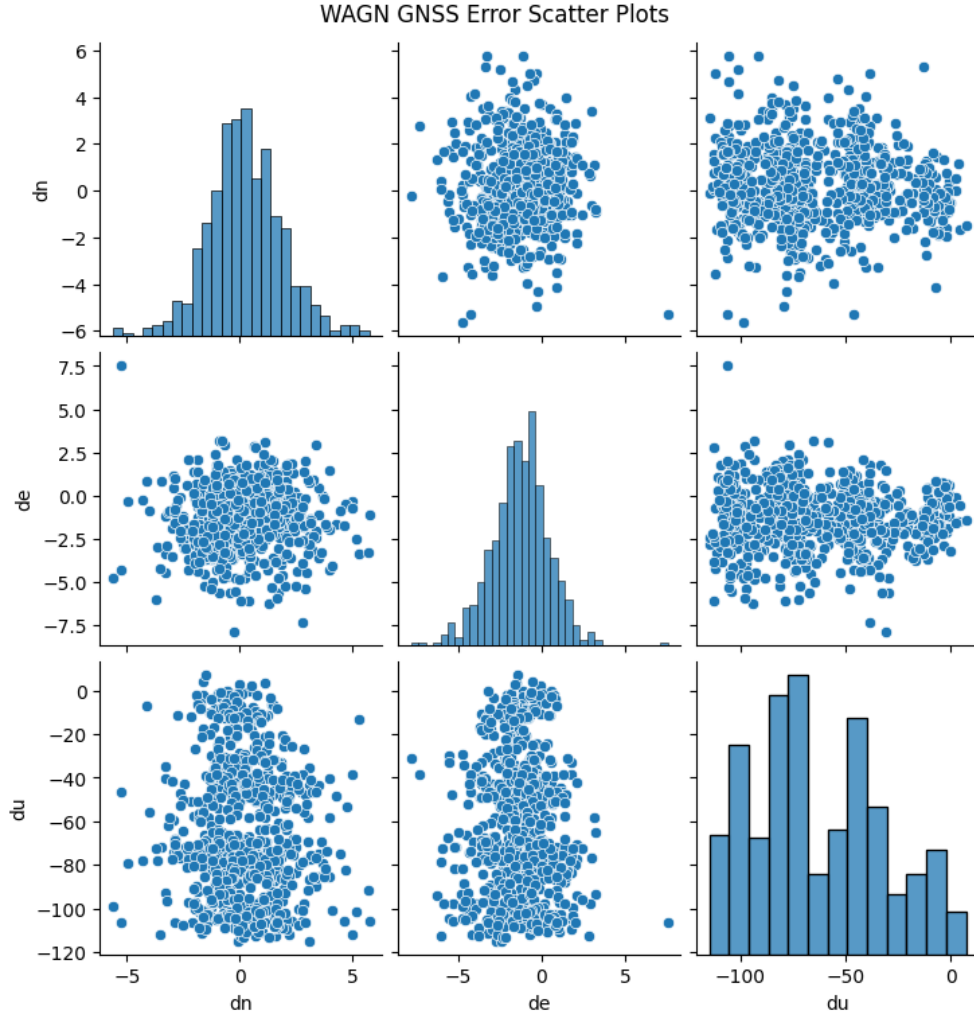


Figure 20: WAGN Scatter Plots

Overall, these low correlations imply that horizontal and vertical positioning errors are mostly uncorrelated for both stations. From a practical standpoint, this indicates that corrective measures or error modeling for one component (e.g.,  $dn$ ) may not directly reduce errors in the other components ( $dE$  or  $dUp$ ). Visualizing these relationships using scatter plots or correlation heatmaps can effectively highlight these patterns. For reporting purposes, a heatmap with color-coded correlation strengths is recommended to give an immediate overview of inter-component dependencies for both ADCN and WAGN.

## 9. Conclusions

The analysis revealed that GNSS positioning errors vary seasonally and by station, with WAGN exhibiting larger vertical deviations than ADCN. Horizontal errors (dN, dE) are relatively small and weakly correlated with vertical errors. Seasonal and meteorological influences are evident, emphasizing the need for targeted error monitoring and correction strategies.